Balancing Evidentiary Value and Sample Size of Adaptive Designs

With Application to Animal Experiments

Leonhard Held
Joint work with Fadoua Balabdaoui und Samuel Pawel

University of Zurich

Boost rigor of animal experiments

ACD WORKING GROUP ON ENHANCING RIGOR, TRANSPARENCY, AND TRANSLATABILITY IN ANIMAL RESEARCH

FINAL REPORT
June 11, 2021

Recommendations to improve study design and analysis

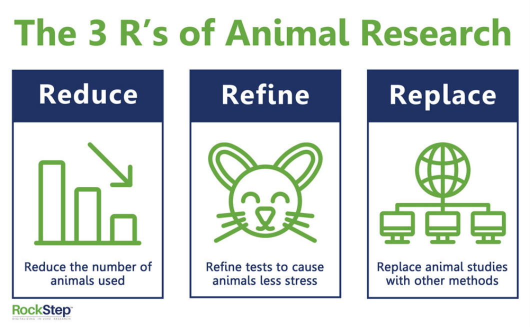
- Improve and expand statistical training for animal researchers
- Facilitate collaboration between statisticians and animal researchers
- Ensure research plans describe elements of study design, analysis, blinding, randomization, etc.
- Evaluate where in pre-research process engagement of statistical expert brings most value

NIH should boost rigor of animal studies with stronger statistics, pilot studies, experts say

By Jocelyn Kaiser | Jun. 16, 2021 , 12:00 PM

Kaiser (2021), Science

Reduce the number of animals used



"[...] to use methods for obtaining comparable levels of information from the use of fewer animals in scientific procedures, or for obtaining more information from the same number of animals"

`swiss3rcc.org/3rs-for-the-public/`

N-hacking

- Also known as peeking, optional stopping
- Uncontrolled analysis of results before the final number of experimental units has been reached $\rightarrow$ Type-I error (T1E) rate inflation

PLOS BIOLOGY

ESSAY

Is *N*-Hacking Ever OK? The consequences of collecting more data in pursuit of statistical significance

Pamela Reinagel *

Department of Neurobiology, School of Biological Science, University of California San Diego, La Jolla, California, United States of America

Reinagel (2023), PLOS Biology

Reinagel's "constrained sample augmentation"

- Interim analyses after $n = 8, 12, \ldots, n_{\texttt{Max}}$ animals, up to $n_{\texttt{Max}} = 32$

Reinagel's "constrained sample augmentation"

- Interim analyses after $n = 8, 12, \ldots, n_{\mathtt{Max}}$ animals, up to $n_{\mathtt{Max}} = 32$
- Stop for success if $p < 0.05$, stop for futility if $p > 0.1$

Reinagel's "constrained sample augmentation"

- Interim analyses after $n = 8, 12, \ldots, n_{\texttt{Max}}$ animals, up to $n_{\texttt{Max}} = 32$
- Stop for success if $p < 0.05$, stop for futility if $p > 0.1$

$\rightarrow$ Moderate T1E rate inflation to 6.25%, improved positive predictive value

Reinagel's "constrained sample augmentation"

- Interim analyses after $n = 8, 12, \ldots, n_{\mathtt{Max}}$ animals, up to $n_{\mathtt{Max}} = 32$
- Stop for success if $p < 0.05$, stop for futility if $p > 0.1$

$\rightarrow$ Moderate T1E rate inflation to 6.25%, improved positive predictive value

$\rightarrow$ How do group-sequential methods perform in this setting?

PERSPECTIVE

Increasing efficiency of preclinical research by group sequential designs

Konrad Neumann[1], Ulrike Grittner[1,2]*, Sophie K. Piper[1,2,3], Andre Rex[2,4], Oscar Florez-Vargas[5], George Karystianis[6], Alice Schneider[1,2], Ian Wellwood[2,7], Bob Siegerink[2,8], John P. A. Ioannidis[9], Jonathan Kimmelman[10], Ulrich Dirnagl[2,3,4,8,11,12]

Neumann et al. (2017), PLOS Biology

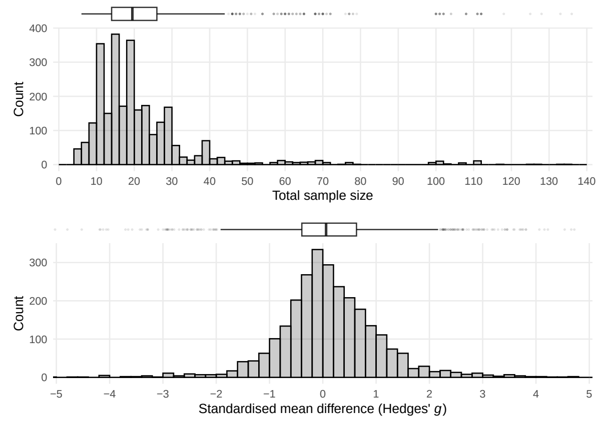
Group sequential designs for in vivo studies: Minimizing animal numbers and handling uncertainty in power analysis

Susanne Blotwijk[a,*], Sophie Hernot[b], Kurt Barbé[a]

Blotwijk et al. (2022), Research in Veterinary Science

Application

Data on 2738 experiments with mice and rats from Neuroscience and Metabolism



Bonapersona et al. (2021), Nature Neuroscience

How many animals could be saved with group-sequential designs?

Simulate z-value z_1 at interim analysis (sample size $n_1 \approx n_2/2$), given the final z-value z_2

$$Z_1 \mid Z_2 = z_2 \sim \mathrm{N}\left(z_2\sqrt{n_1/n_2}, 1 - n_1/n_2\right)$$

to assess the effect of early stopping:

Method	Mean n	H_0 rejections (%)	Interim efficacy stops (%)	Interim futility stops (%)	Animals saved
No interim analysis	22.5	27.8	0	0	0
Haybittle-Peto	21.2 (21.1, 21.3)	28.3 (28, 28.5)	11.8 (11, 12.5)	0 (0, 0)	3'804 (3'515, 4'099)
O'Brien-Fleming	21.4 (21.3, 21.5)	27.6 (27.4, 27.8)	9.8 (9.1, 10.4)	0 (0, 0)	3'152 (2'891, 3'426)
Pocock	20.6 (20.5, 20.7)	25.9 (25.3, 26.4)	16.7 (15.9, 17.6)	0 (0, 0)	5'384 (5'043, 5'738)
Haybittle-Peto + futility	18.2 (18, 18.4)	27.6 (27.2, 28)	11.8 (11, 12.5)	28.7 (27.2, 30.2)	11'858 (11'263, 12'470)
O'Brien-Fleming + futility	18.4 (18.2, 18.6)	26.9 (26.6, 27.2)	9.8 (9.1, 10.4)	29.4 (27.9, 31)	11'423 (10'832, 12'024)
Pocock + futility	16.7 (16.5, 17)	25.3 (24.7, 25.9)	16.7 (15.9, 17.6)	37.1 (35.6, 38.7)	15'905 (15'286, 16'541)
Reinagel	12.4 (12.3, 12.5)	23.6 (22.6, 24.6)	20.4 (19.4, 21.4)	72.7 (71.5, 73.8)	27'771 (27'416, 28'110)

Additional futility stopping based on predictive power $\leq 10\%$.

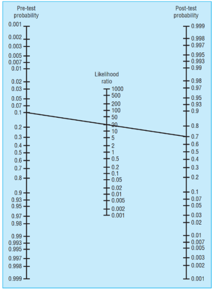
Talk structure

1. Adopt diagnostic likelihood ratios and diagnostic odds ratio to hypotheses testing
2. Incorporate sample size to define Experimental Unit Information Index (EUII), the average information gain from one animal
3. Extend formulation to adaptive designs
4. Compare different adaptive designs in simulation study

Likelihood ratios

- Likelihood ratios (LRs) quantify the change in odds after having observed a positive (significant, $+$) or negative (not significant, $-$) test result:

$$\text{Posterior odds}(H_1) = \left\{ \begin{matrix} \text{LR}^{+} \\ \text{LR}^{-} \end{matrix} \right\} \cdot \text{Prior odds}(H_1)$$



Use of Fagan's nomogram for calculating post-test probabilities[7]

Statistics Notes

Diagnostic tests 4: likelihood ratios

Jonathan J Deeks, Douglas G Altman

Deeks and Altman (2004), BMJ

Likelihood ratios for hypotheses testing

Contents lists available at ScienceDirect
Journal of Mathematical Psychology
journal homepage: www.elsevier.com/locate/jmp

Rejection odds and rejection ratios: A proposal for statistical practice in testing hypotheses

M.J. Bayarri[a,†], Daniel J. Benjamin[b], James O. Berger[c,*], Thomas M. Sellke[d]

Received: 30 March 2022 | Revised: 11 November 2022 | Accepted: 15 November 2022
DOI: 10.1002/bimj.202200102
Biometrical Journal
RESEARCH ARTICLE

Relative likelihood ratios for neutral comparisons of statistical tests in simulation studies

Qiuxi Huang[1] | Ludovic Trinquart[1,2,3]

positive LR (rejection ratio): $\mathrm{LR}^{+} = \dfrac{\text{Power}}{\text{T1E rate}} \left(e.g. \dfrac{80\%}{5\%} = 16\right)$

Likelihood ratios for hypotheses testing

Contents lists available at ScienceDirect

Journal of Mathematical Psychology

journal homepage: www.elsevier.com/locate/jmp

ELSEVIER

Rejection odds and rejection ratios: A proposal for statistical practice in testing hypotheses

M.J. Bayarri[a,†], Daniel J. Benjamin[b], James O. Berger[c,*], Thomas M. Sellke[d]

Received: 30 March 2022 | Revised: 11 November 2022 | Accepted: 15 November 2022

DOI: 10.1002/bimj.202200102

Biometrical Journal

RESEARCH ARTICLE

Relative likelihood ratios for neutral comparisons of statistical tests in simulation studies

Qiuxi Huang[1] | Ludovic Trinquart[1,2,3]

positive LR (rejection ratio): $\mathrm{LR}^+ = \dfrac{\text{Power}}{\text{T1E rate}} \left(e.g.\ \dfrac{80\%}{5\%} = 16\right)$

negative LR (nonrejection ratio): $\mathrm{LR}^- = \dfrac{1-\text{Power}}{1-\text{T1E rate}} \left(e.g.\ \dfrac{20\%}{95\%} = 4/19\right)$

Likelihood ratios for hypotheses testing

Contents lists available at ScienceDirect

Journal of Mathematical Psychology

journal homepage: www.elsevier.com/locate/jmp

ELSEVIER

Rejection odds and rejection ratios: A proposal for statistical practice in testing hypotheses

M.J. Bayarri [a,†], Daniel J. Benjamin [b], James O. Berger [c,*], Thomas M. Sellke [d]

Received: 30 March 2022 | Revised: 11 November 2022 | Accepted: 15 November 2022

DOI: 10.1002/bimj.202200102

Biometrical Journal

RESEARCH ARTICLE

Relative likelihood ratios for neutral comparisons of statistical tests in simulation studies

Qiuxi Huang[1] | Ludovic Trinquart[1,2,3]

positive LR (rejection ratio): $\mathrm{LR}^{+} = \dfrac{\text{Power}}{\text{T1E rate}} \left(e.g.\ \dfrac{80\%}{5\%} = 16\right)$

negative LR (nonrejection ratio): $\mathrm{LR}^{-} = \dfrac{1 - \text{Power}}{1 - \text{T1E rate}} \left(e.g.\ \dfrac{20\%}{95\%} = 4/19\right)$

Diagnostic odds ratio: $\mathrm{DOR} = \mathrm{LR}^{+}/\mathrm{LR}^{-} \ (e.g.\ 16/(4/19) = 76)$

Glas et al. (2003), J Clin Epi

Bayesian Updating with Likelihood Ratios

Prevalence

Prior Odds
1 to 4
(20% probability)

Hypothesis Test Result
(Likelihood Ratio)

significant
16 to 1

not significant
4 to 19

Predictive Value

Posterior Odds

4 to 1
(80% probability)

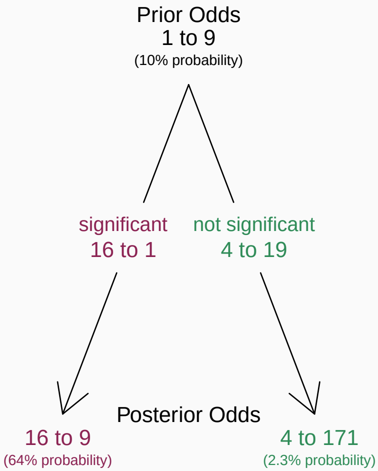
1 to 19
(5% probability)

Bayesian Updating with Likelihood Ratios

Prevalence

Hypothesis Test Result
(Likelihood Ratio)

Predictive Value



Diagnostic odds ratios for hypotheses testing

Important reformulations:

1. Frequentist:

$$\boxed{\text{DOR} = \frac{\text{Power Odds}}{\text{T1E Odds}}}$$

Diagnostic odds ratios for hypotheses testing

Important reformulations:

1. Frequentist:

$$\boxed{\mathrm{DOR} = \frac{\text{Power Odds}}{\text{T1E Odds}}}$$

2. Bayesian:

$$\boxed{\mathrm{DOR} = \frac{\mathrm{Odds}(H_1 \mid \text{significant})}{\mathrm{Odds}(H_1 \mid \text{not significant})}}$$

Experimental unit-information index (EUII)

- Motivation: Likelihood ratios act multiplicatively on the odds of H_1, so the average contribution from one experimental unit is $\text{LR}^{1/n}$:

$$\text{LR} = \underbrace{\text{LR}^{1/n} \times \text{LR}^{1/n} \times \ldots \times \text{LR}^{1/n}}_{n \text{ times}}$$

$\rightarrow$ $\text{DOR}^{1/n}$ represents the evidentiary value of one experimental unit:

$$\boxed{\text{EUII} = \frac{(\text{LR}^{+})^{1/n}}{(\text{LR}^{-})^{1/n}} = \text{DOR}^{1/n}}$$

One-sample z-test

Power (in %)	Type-I error rate (in %)	DOR	n	EUII
80	5.0	76	31.4	1.15
80	2.5	156	38.0	1.14
90	5.0	171	42.0	1.13
90	2.5	351	49.6	1.13

$\uparrow$

n: Required sample size to detect a standardised effect of $\delta = 0.5$

One-sample z-test

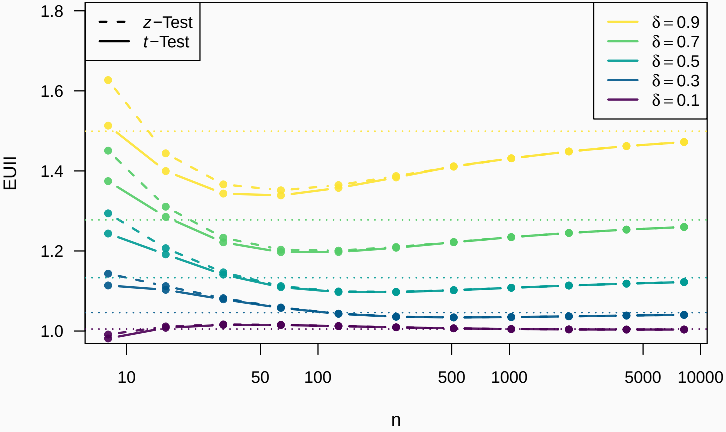
Power (in %)	Type-I error rate (in %)	DOR	n	EUII
80	5.0	76	31.4	1.15
80	2.5	156	38.0	1.14
90	5.0	171	42.0	1.13
90	2.5	351	49.6	1.13

$\uparrow$

n: Required sample size to detect a standardised effect of $\delta = 0.5$

Note: $\exp(\delta^2/2) = 1.133$

EUII for increasing n and fixed $\alpha = 5\%$



Limiting value $\lim_{n \to \infty} \text{EUII} = \exp\left(\delta^2/2\right)$

EUII for adaptive designs

- Sample size N is now a random variable
- Distinguish sample size for significant (N_+) and not significant (N_-) findings:

$$\text{EUII} = \frac{\left(\text{LR}^+\right)^{\text{E}(1/N_+)}}{\left(\text{LR}^-\right)^{\text{E}(1/N_-)}} \approx \frac{\left(\text{LR}^+\right)^{1/\,\text{E}(N_+)}}{\left(\text{LR}^-\right)^{1/\,\text{E}(N_-)}}$$

Computation of $\text{E}(N_+)$ and $\text{E}(N_-)$ requires specification of prior probability $\Pr(H_1)$:

$$\begin{aligned}
\text{E}(N_+) &= \text{E}(N_{0+}) \Pr(H_0 \mid \text{significant}) \\
&+ \text{E}(N_{1+}) \Pr(H_1 \mid \text{significant}) \\
\text{E}(N_-) &= \text{E}(N_{0-}) \Pr(H_0 \mid \text{not significant}) \\
&+ \text{E}(N_{1-}) \Pr(H_1 \mid \text{not significant})
\end{aligned}$$

	Test result		
Hypothesis	significant	not significant	
H_0	$\text{E}(N_{0+})$	$\text{E}(N_{0-})$	$\text{E}(N_0)$
H_1	$\text{E}(N_{1+})$	$\text{E}(N_{1-})$	$\text{E}(N_1)$
	$\text{E}(N_+)$	$\text{E}(N_-)$	$\text{E}(N)$

Second order approximation

Coefficient of variation

$$\mathrm{CV}(N_+) = \frac{\sqrt{\mathrm{Var}(N_+)}}{\mathrm{E}(N_+)} \qquad \mathrm{CV}(N_-) = \frac{\sqrt{\mathrm{Var}(N_-)}}{\mathrm{E}(N_-)}$$

is needed for second-order Taylor approximation

$$\mathrm{E}(1/N_+) \approx \frac{1 + \mathrm{CV}(N_+)^2}{\mathrm{E}(N_+)} \qquad \mathrm{E}(1/N_-) \approx \frac{1 + \mathrm{CV}(N_-)^2}{\mathrm{E}(N_-)}$$

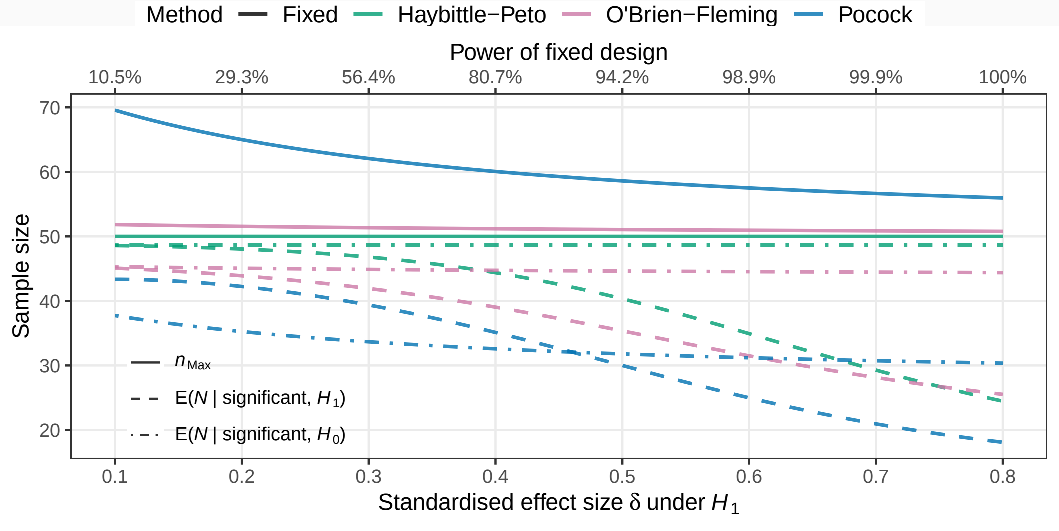
$$\rightarrow \quad \widetilde{\mathrm{EUII}} = \frac{(\mathrm{LR}^+)^{(1+\mathrm{CV}(N_+)^2)/\,\mathrm{E}(N_+)}}{(\mathrm{LR}^-)^{(1+\mathrm{CV}(N_-)^2)/\,\mathrm{E}(N_-)}}$$

Characteristics of group-sequential designs

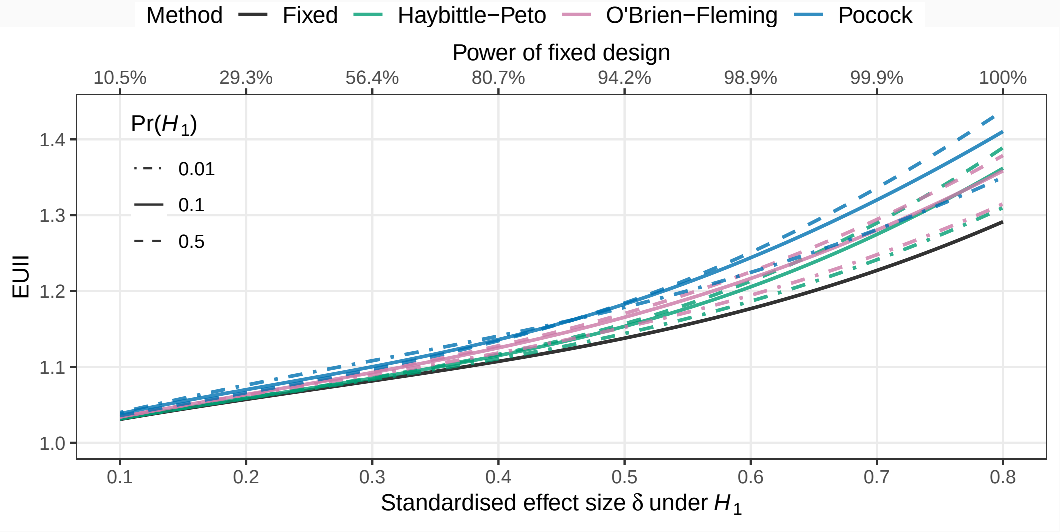
Method	a_1	a_2	a_3	a_4	T1E rate (in %)	$n_{\mathtt{Max}}$
Pocock	0.00911	0.0091	0.0091	0.0091	2.50	59.2
O'Brien-Fleming	0.00003	0.0021	0.0097	0.0215	2.50	51.1
Haybittle-Peto	0.00050	0.0005	0.0005	0.0250	2.54	50.0

Comparison with fixed design ($n = 50$)

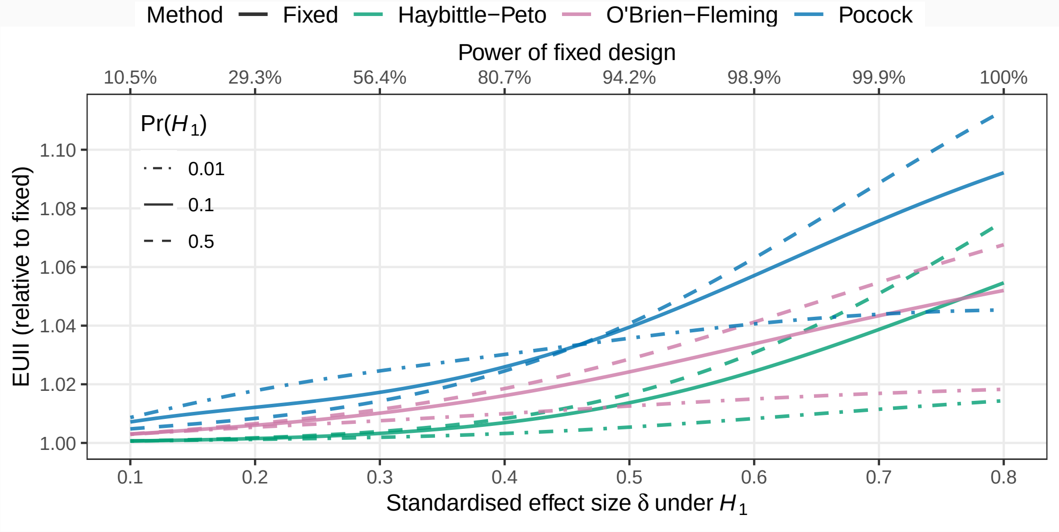
Comparison of group-sequential designs: Expected sample size



Comparison of group-sequential designs: EUII



Comparison of group-sequential designs: relative EUII



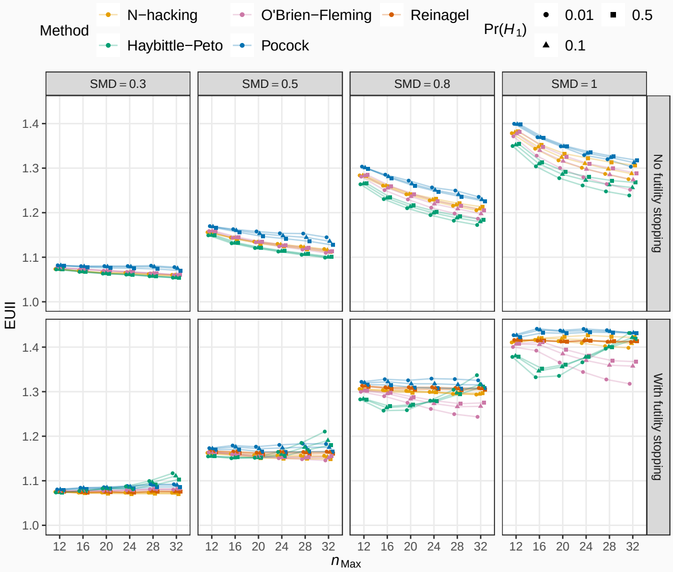
Simulation study

- Compare Reinagel approach with three group-sequential methods:
    - Pocock
    - O'Brien-Fleming
    - Haybittle-Peto (uncontrolled)

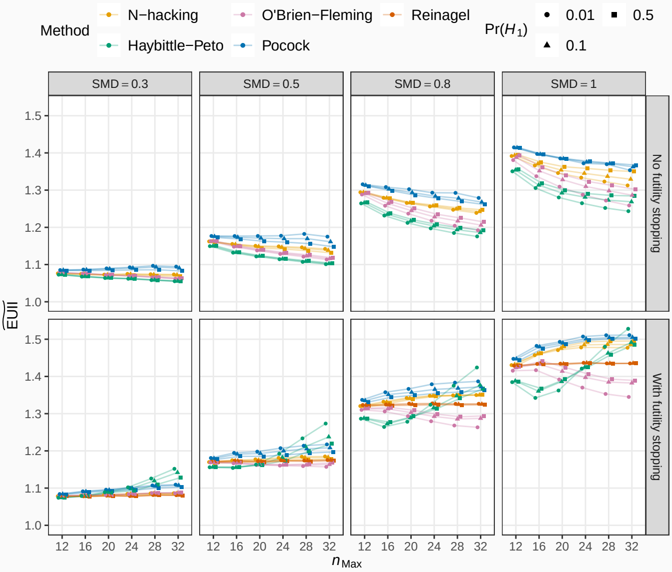
  for different values of

    - $n_{\texttt{Max}}$
    - δ (standardised mean difference, SMD)
    - $\Pr(H_1)$
- Additional stopping for futility based on predictive power < 10% at the next interim analysis
- Use first and second order approximation in EUII computation

EUII of several group-sequential methods (1st order approximation)



EUII of several group-sequential methods (2nd order approximation)



Discussion

A new measure of statistical information that takes into account Type-I error rate, power, and sample size

- Future applications:
  - Bayesian sequential designs
  - Dual-criterion
  - More complex experimental (block) designs
- Average the power over plausible values of the effect size
- Monte Carlo standard error of EUII

More Details

Balancing Evidentiary Value and Sample Size of Adaptive Designs with Application to Animal Experiments

Leonhard Held[1]*, Fadoua Balabdaoui[2] Samuel Pawel[1]

References

Blotwijk, S., Hernot, S., and Barbé, K. (2022). Group sequential designs for in vivo studies: Minimizing animal numbers and handling uncertainty in power analysis. *Research in Veterinary Science*, 145:248–254. doi:10.1016/j.rvsc.2022.03.003.

Bonapersona, V., Hoijtink, H., Sarabdjitsingh, R. A., and Joëls, M. (2021). Increasing the statistical power of animal experiments with historical control data. *Nature Neuroscience*, 24(4):470–477. doi:10.1038/s41593-020-00792-3.

Deeks, J. J. and Altman, D. G. (2004). Diagnostic tests 4: likelihood ratios. *BMJ*, 329:168–169.

Glas, A. S., Lijmer, J. G., Prins, M. H., Bonsel, G. J., and Bossuyt, P. M. (2003). The diagnostic odds ratio: a single indicator of test performance. *Journal of Clinical Epidemiology*, 56(11):1129–1135. doi:10.1016/S0895-4356(03)00177-X.

Kaiser, J. (2021). NIH should boost rigor of animal studies with stronger statistics, pilot studies, experts say. *Science*. doi:10.1126/science.abk0214.

Neumann, K., Grittner, U., Piper, S. K., Rex, A., Florez-Vargas, O., Karystianis, G., Schneider, A., Wellwood, I., Siegerink, B., Ioannidis, J. P. A., Kimmelman, J., and Dirnagl, U. (2017). Increasing efficiency of preclinical research by group sequential designs. *PLOS Biology*, 15(3):e2001307. doi:10.1371/journal.pbio.2001307.

Reinagel, P. (2023). Is N-Hacking Ever OK? The consequences of collecting more data in pursuit of statistical significance. *PLOS Biology*, 21(11):e3002345. doi:10.1371/journal.pbio.3002345.